Tanvir Hossain Ovi

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EDUCATION

University of Chittagong, Bangladesh *Jan 2020 – Aug 2025*
BSc in Electrical and Electronic Engineering | **Engineering Degree with Honours**
CGPA: 3.78 / 4.00 | **Merit Position: 2nd out of 63 students**
Thesis: *EEG-based cognitive load analysis in 360° virtual reality and 2D laptop video modes*

RESEARCH INTERESTS

Brain-computer interfaces • Neural decoding • EEG/iEEG signal analysis • Cross-subject generalization

PUBLICATIONS

Ten papers: four journal articles (three published, one accepted), one presented IEEE conference paper, and five under review or submitted. First author on six.

Journal Articles

[P1] A. Hossain, M. A. Z. Akas, **T. H. Ovi**, and M. F. Kader, "Heterogeneous Deep Learning Approaches for EEG-Based Imagined Bangla Word-Level Speech Recognition," *Computer Speech & Language*, accepted for publication, 2026.

[P2] **T. H. Ovi**, M. M. Islam, M. A. Uddin, K. Ghazinour, A. M. M. Chowdhury, and N. Siddique, "Frequency Band Dissociation Between Cognitive State Recognition and Person Identification in EEG," *Machine Learning with Applications*, p. 100901, 2026.

[P3] M. A. Z. Akas, **T. H. Ovi**, and M. F. Kader, "Impact of Mobile Phone Use on the Brain Activity: Audio Call vs Video Call," *Acta Psychologica*, vol. 262, p. 106160, 2026.

[P4] A. Hossain, **T. H. Ovi**, M. A. I. Mahmood, and M. F. Kader, "Classification of Envisioned English Speech from EEG using Deep Learning Approaches," *Machine Learning with Applications*, vol. 22, p. 100752, 2025.

Conference Paper

[P5] **T. H. Ovi**, A. M. M. Chowdhury, K. Ghazinour, and M. Simsek, "Design of a Clustering-Enabled Multi-Stage Intrusion Detection Framework for APT Mitigation in IoT Networks," *IEEE Conference on Dependable and Secure Computing (DSC)*, 2026. [Presented]

Under Review and Submitted

[P6] **T. H. Ovi**, T. Forsythe, A. M. M. Chowdhury, N. Siddique, and Y. Li, "Phenotype-Aware Personalization for Cross-Session EEG-Based Emotion Recognition," *Expert Systems With Applications*. [Under Review]

[P7] **T. H. Ovi**, A. M. M. Chowdhury, S. Shubbar, K. Watanabe, and N. Siddique, "Complementary Raw and Engineered Representations for Focal and Non-Focal Intracranial EEG Classification under a Recording-Separated Split," *Knowledge-Based Systems*. [Submitted]

[P8] **T. H. Ovi**, T. F. Arin, M. A. I. Mahmood, and M. F. Kader, "EEG-Based Cognitive Load Analysis in Immersive Virtual Reality and Laptop-Based Video Learning Environments," *Applied Cognitive Psychology*. [Submitted]

[P9] **T. H. Ovi**, A. M. M. Chowdhury, and N. Siddique, "E-SpecID: Electrode-Wise Spectral Learning for Cross-Session EEG Person Identification," *Brain Theory and Intelligence Science*. [Submitted]

[P10] T. F. Arin, **T. H. Ovi**, and M. F. Kader, "Bangla Imagined Speech Using EEG to Understand Emotional Words," *Acta Psychologica*. [Submitted]

RESEARCH EXPERIENCE

Research Assistant, Biometric Intelligence and Vision Laboratory *Jul 2025 – Present*
Department of Cybersecurity, State University of New York, Canton, USA | Remote

Supervised by Dr. A M Mahmud Chowdhury and co-supervised by Dr. Nazmul Siddique (Ulster University, UK). First author on all five studies from the appointment [P2, P5, P6, P7, P9].

- **Intracranial EEG classification** [P7]. Traced inflated Bern-Barcelona accuracies to window-level splits, rebuilt the evaluation at recording level, and reached 96.27% with CSRA-EF Net.
- **Cross-session emotion recognition** [P6]. Placed 80.1% of explained variance in the participant-by-emotion interaction. Calibration with phenotype pooling raised accuracy from 39.96% to 91.92%.
- **Frequency band dissociation** [P2]. Delta EEG reached 86.9% task accuracy but only 15.6 to 23.6% authentication, while broadband reached 69.9%.

Undergraduate Researcher, University of Chittagong *2023 – 2025*
Department of Electrical and Electronic Engineering, Chattogram, Bangladesh

- Recorded EEG from participants for four studies: the VR thesis, imagined Bangla words, emotionally valenced Bangla words, and audio vs. video calls [P1, P3, P8, P10].
- **BSc thesis** [P8]. Designed a VR vs. laptop video protocol and recorded 30 participants on a 14-channel Emotiv EPOC X. Cognitive load (theta-to-alpha ratio) was significantly lower in the VR group.

HONORS AND AWARDS

- First Runner-up, Student Merit Award, EEE Fest, University of Chittagong *Feb 2025*
- Champion, Robo Soccer Competition, EEE Fest, University of Chittagong *Jan 2023*
- Second Runner-up, Robo Soccer Competition, Engineering Day, University of Chittagong *Jun 2022*

TEST SCORES

IELTS Academic *14 April 2026*
Overall Band Score: **7.5** | Listening: 8.0 Reading: 8.0 Writing: 6.0 Speaking: 7.0

TECHNICAL SKILLS

Programming	Python, PyTorch, TensorFlow, Keras, scikit-learn, Matplotlib, Git, Linux, LaTeX
EEG	Emotiv EPOC X recording, MNE-Python, EEGLAB, ICA artifact removal
Statistics	t-test, Wilcoxon, ANOVA, MANOVA, FDR correction, hierarchical clustering

REFERENCES

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